

# Tiphaine Delaunay

📍 LMV, Bâtiment Fermat, 45 av. des États-Unis, 78000 Versailles, France

🔗 <https://tiphainedelaunay.github.io>

## Research Experience

|                                                                                                                                                                                                                                                                                                                                                                                             |                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| <b>Maîtresse de conférences (Assistant Professor)</b><br>LMV, Université de Versailles Saint-Quentin-en-Yvelines                                                                                                                                                                                                                                                                            | Since Sept. 2026          |
| <b>Postdoctoral Researcher</b><br>LAGA, Sorbonne Paris Nord University<br><i>Reconstruction of relative Lamé parameters in linear elasticity</i><br>With M. Darbas (LAGA, Sorbonne Paris Nord University) and J. Heleine (IMT, Paul Sabatier University, Toulouse)<br>Funded by ANR DynImplant                                                                                              | Sept. 2025 – August 2026  |
| <b>Postdoctoral Researcher</b><br>MONC, Inria & Institute of Mathematics of Bordeaux<br><i>Deciphering tumor response to propranolol in angiosarcomas by mathematical modeling and data assimilation</i><br>Project led by A. Collin (MACS, Nantes University) and F. Moisan (BRIC)<br>With C. Etchegaray (MONC, Inria), F. Laanani (BRIC), S. Oucherif (BRIC) and L. Pechtimaldjian (BRIC) | February 2024 – June 2025 |
| <b>PhD in Applied Mathematics</b><br>MEDISIM, Inria & École Polytechnique, Institut Polytechnique de Paris - CNRS<br><i>Adaptive observers for the wave equation and associated discretizations: formulations and analysis</i><br>Supervised by P. Moireau (ANANKE, École Polytechnique) and S. Imperiale (ANANKE, Inria)<br>Funded by ANR ODISSE, <a href="#">Link to manuscript</a>       | 2020 – 2023               |

## Education

|                                                                                                                                                                                              |             |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>PhD Degree in Applied Mathematics</b><br>MEDISIM, Inria & École Polytechnique, Institut Polytechnique de Paris - CNRS                                                                     | 2020 – 2023 |
| <b>Engineering Degree</b><br>INSA Rouen Normandie, Department of Mathematical Engineering                                                                                                    | 2015 - 2020 |
| <b>Research Master's Degree</b><br>University of Rouen Normandy, Specialization in Pure and Applied Mathematics<br>Thesis: 3D reconstruction from two-dimensional slices applied to medicine | 2019 - 2020 |

## Mathematical interests

- Inverse problems and data assimilation
- Mathematical modelling with PDEs
- Evolution problem
- Waves
- Biomedical applications

## Publications

---

Gradient-based reconstruction method of relative elastic material parameters from full-field measurements

M. Darbas, **T. Delaunay**, J. Heleine (2026)

Submitted, [HAL Link](#)

Modeling and Parameter Estimation of Tumor Spheroids from Bottom-View Imaging: Beyond the Spherical Assumption

A. Collin, **T. Delaunay**, C. Etchegaray, F. Laanani, F. Moisan, S. Oucherif, L. Pechtimaldjian (2026)

Submitted, [HAL Link](#)

Solving inverse source wave problem - From Carleman estimates to observer design

M. Boulakia, M. de Buhan, **T. Delaunay**, S. Imperiale, P. Moireau (2025)

*Mathematical Control and Related Fields*, DOI: [10.1007/s00211-024-01440-9](#), [HAL Link](#)

Mathematical analysis of an observer for solving inverse source wave problem

**T. Delaunay**, S. Imperiale, P. Moireau (2025)

*Inverse problem and imaging*, DOI: [10.3934/ipi.2025043](#), [HAL Link](#)

Uniform boundary stabilization of a high-order finite element space discretization of the 1-d wave equation

**T. Delaunay**, S. Imperiale, P. Moireau (2024)

*Numerische Mathematik*, DOI: [10.1007/s00211-024-01440-9](#), [HAL Link](#)

Adaptative observers for wave equations and associated discretizations : formulations and analysis

**T. Delaunay** (2023), [HAL Link](#)

## Teaching

---

### Mathematics for Engineers

2025

Sup Galilée, Villetaneuse, France, (Practical class L3) - 39h

### Tutored interdisciplinary project

2024

Master Cancer Biology of University of Bordeaux, Talence, France, (Project M2) - 5h

Creation and supervision of a project: *Mathematical modeling of the growth of spheroids in free growth and in responses to a non-cytotoxic drug*

### Introduction to PDEs and finite differences

2022 & 2023

ENSTA Paris, Palaiseau, France, (Practical class L3) - 2 x 14h

### Finite element method

2021 & 2022

ENSTA Paris, Palaiseau, France, (Practical class M1) - 2 x 14h

## Science popularization

---

- Led a cryptography workshop during the Institut Polytechnique de Paris Science Fair, 2021
- Created a popularization science video on mathematical modeling and numerical simulation for middle and high school students, 2021, [URL](#)
- Led a two-day research project for five middle school girls on Conway sequences in the "Rendez-vous des jeunes mathématiciennes et informaticiennes", online, 2021
- Popularization conference in a High School on cryptography, Académie de Créteil, 2021

## Oral Presentations

---

### *Talks at national and international conferences*

|                                                                                                                                               |      |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>PICOF</b><br>Hammamet, Tunisia, <a href="#">URL</a>                                                                                        | 2025 |
| <b>12ème Biennale de la Société des Mathématiques Appliquées et Industrielles (SMAI)</b><br>Carcans-Maubuisson, France, <a href="#">URL</a>   | 2025 |
| <b>Rencontres Normandes sur les aspects théorique et numérique des EDP</b><br>Saint-Etienne-du-Rouvray, France, <a href="#">URL</a>           | 2024 |
| <b>WAVES</b><br>Palaiseau, France, <a href="#">URL</a>                                                                                        | 2022 |
| <b>European Congress on Computational Methods in Applied Sciences and Engineering (ECCOMAS)</b><br>Oslo, Norway, <a href="#">URL</a>          | 2022 |
| <b>Rencontre Jeunes Chercheuses, Jeunes Chercheurs : Ondes</b><br>Inria Université Côte d'Azur, Sophia Antipolis, France, <a href="#">URL</a> | 2022 |
| <b>Rencontre pour les Ondes et leurs applications</b><br>INSA Rouen Normandie, Saint-Etienne-du-Rouvray, France, <a href="#">URL</a>          | 2022 |
| <b>Congrès des Jeunes Chercheurs en Mathématiques Appliquées</b><br>Palaiseau, France, <a href="#">URL</a>                                    | 2021 |

### *Seminar talks*

|                                                                                                                                                                           |      |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Seminar on mathematical modeling in life sciences and health</b><br>Laboratoire Jacques-Louis Lions, Paris, France, <a href="#">URL</a>                                | 2026 |
| <b>Partial Differential Equations Seminar, IRMA</b><br>Institut de Recherche Mathématique Avancée (IRMA), Strasbourg, France, <a href="#">URL</a>                         | 2026 |
| <b>MAC Seminar</b><br>Institute of Mathematics of Toulouse, Toulouse, France, <a href="#">URL</a>                                                                         | 2026 |
| <b>Applied Mathematics Seminar</b><br>Laboratoire de Mathématiques Jean Leray, Nantes Université, Nantes, France, <a href="#">URL</a>                                     | 2025 |
| <b>Math, Biology and Imaging Seminar</b><br>LAGA, Sorbonne Paris Nord University, Villetaneuse, France, <a href="#">URL</a>                                               | 2025 |
| <b>COMMEDIA Team Seminar</b><br>Laboratoire Jacques-Louis Lions, Paris, France, <a href="#">URL</a>                                                                       | 2024 |
| <b>PDE Seminar of Nancy</b><br>Institut Elie Cartan de Lorraine, Nancy, France                                                                                            | 2024 |
| <b>Seminar on the Mathematical and Statistical Foundation of Future Data-Driven Engineering</b><br>Isaac Newton Institute, Cambridge, United Kingdom, <a href="#">URL</a> | 2023 |
| <b>Workshop on Assimilation, Control and Computational Speedup</b><br>LAGA, Sorbonne Paris Nord University, Villetaneuse, France, <a href="#">URL</a>                     | 2023 |

### *Poster*

|                                                                 |             |
|-----------------------------------------------------------------|-------------|
| <b>IDEFIX-MEDISIM-POEMS Joint Meetings</b><br>Palaiseau, France | 2020 & 2021 |
|-----------------------------------------------------------------|-------------|